

TRU
KIRAN
INTERNATIONAL COLLEGE OF
ENGINEERING & MANAGEMENT

AFFILIATED
TRIBHUVAN UNIVERSITY

Mid Term Examination – 2025 (2081)

SET B

Semester

Subject: CACS 203: System Analysis & Design

F.M.: 60

P.M.: 24

Time: 3hrs.

Candidates are required to answer the questions in their own words as far as possible. Figure in the margin indicates full marks. [6X5=30]

Attempt Any SIX Questions:

Group B:

11. Define Project Planning? Explain major activities of project planning process.
12. Explain concept of IS Development Planning Process in reference with Corporate Strategic Planning Process.
13. What is information system? Explain different types of Information Systems.
14. What are the steps of Identifying and Selecting IS Development Projects? Explain.
15. What is Project Scope Statement? Explain its major components.
16. Identify the actor and use cases for Examination Management system of TU and draw a use case diagram.
17. Explain Business Process Reengineering (BPR) as agile approach of requirement determination.
18. Explain Importance of PERT in project planning with suitable numerical example.

Attempt Any TWO Questions:

[2X10=20]

Group C:

19. Assuming Monetary benefits of an information system at Rs 70000 per year, onetime costs of Rs 80000, recurring costs of Rs 30000 per year, a discount rate of 11 % and five year time horizons, calculate the following:
 - a. Net Present value of the costs and Benefits [3]
 - b. The overall return on the investment [3]
 - c. The point at which breakeven occur [4]
20. Explain concept of Use case and Prototyping approach of requirement determination process [5+5]
21. Define CASE tool. Explain Different types of CASE Tools. How DFD can be used in process modeling of system to be developed? Explain Different symbols used in DFD. [2+4+4]

What will do
why will do
when will do



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Mid Term Examination – 2025 (2081)

SET B

3rd Semester

Subject: CACS 205: Web Technology

Time: 3hrs.

F.M.: 60

P.M.: 24

Candidates are required to answer the questions in their own words as far as possible. Figure in the margin indicates full marks.

Group B:

Attempt Any TWO Questions:

[2X10=20]

11. What is the role of CSS while designing the web pages? Describe the ways of embedding CSS in html document.
12. What is responsive website? Why responsive website is necessary these days? How can you create the responsive website, explain with examples?
13. What is the significance of HTML forms? Explain any five types of form input fields with example.

Group C:

Attempt Any SIX Questions:

[6X5=30]

14. Define front-end development and back-end development of the website. What is the importance of server-side scripting language?
15. Why PHP is popular among the web developer?
16. What is Tier Technology? Explain 3 tier technology with examples.
17. Differentiate between GET and POST method with examples.
18. Create a form to enter the user's name and email. Write PHP code to display these information in new page.
19. Compare between radio button and dropdown in html. Explain with example.
20. What is imagemap? Write code to illustrate its importance.

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Mid Term Examination – 2025 (2081)

SET A

BCA, 3rd Semester

Subject: CACS 201: Data Structure and Algorithms

Time: 3hrs.

F.M.: 60

P.M.: 24

Candidates are required to answer the questions in their own words as far as possible. Figure in the margin indicates full marks.

Group B:

Attempt Any SIX Questions:

[6X5=30]

11. What is Data Structure? Explain different operations to be performed on data structure. [1+4]
12. What is Stack? Write an algorithm to push an item in Stack. [2+3]
13. Explain Stack as an Abstract data type. [5]
14. Write an algorithm to convert an infix expression into prefix expression. [5]
15. What is circular queue? Write an algorithm to insert an item in circular queue. [2+3]
16. What are the applications of Queue? Write an algorithm to delete an item from linear queue. [2+3]
17. Define dynamic list. Explain Array implementation of list. [2+3]

Group C:

[2X10=20]

18. Convert following infix expression to postfix showing stack status after every step. $((F + G) - C * H / I) * (A - B) + E$ and evaluate the obtained postfix expression with following values: $A = 5, B = 2, C = 4, D = 2, E = 5, F = 1, G = 2, H = 7, I = 3$. [5+5]
19. What is double linked list? How does it differ from singly linked list? Write an algorithm or function to add a node at the beginning and end of singly linked list. [1+1+4]



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Mid Term Examination – 2025 (2081)

SET A

BCA, 3rd Semester

Subject: CACS 204: OOP in Java

Time: 3hrs.

F.M.: 60

P.M.: 24

Candidates are required to answer the questions in their own words as far as possible. Figure in the margin indicates full marks.

Group B:

Attempt Any SIX Questions:

[6X5=30]

11. Explain the architecture of the Java Virtual Machine (JVM) in detail. How the process of compilation and interpretation done in Java
12. Write steps for Compiling and Running Simple Java Program
13. What is identifier in java? Discuss the rules and conventions for naming identifiers in Java.
14. Discuss the different types of inner class in Java.
15. Define thread. Explain the life cycle of thread.
16. Explain different types of control statements used in java.
17. Short Notes (Any TWO):
 - a. JVM
 - b. Extend keyword
 - c. RandomAccessFile

Group C:

Attempt Any TWO Questions:

[2X10=20]

18. A) What is polymorphism? How polymorphism is achieved using interface.
B) Differentiate abstract class and interface.
19. A) What is Encapsulation. How can we achieve encapsulation.
B) Define Stream. Write a program in java to copy the content from one file to another.
20. A) What is Object Serialization and Deserialization. Write a java program to demonstrate it.
B) Write a Java program that will read balance and withdraw amount from keyboard and display the remaining balance on screen if balance is greater than withdraw amount otherwise throw an User defined exception with appropriate message.

BCA, 3rd Semester

Subject: CAST 202: Probability & Statistics
Time: 3hrs.

F.M.: 60

P.M.: 24

Candidates are required to answer the questions in their own words as far as possible. Figure in the margin indicates full marks.

Group B: Attempt Any SIX Questions: [6X5=30]

11. Median income of 150 workers of a factory is Rs. 7,000. Find the missing frequencies and compute mean income of the workers.

Income in '000' Rs.	0-2	2-4	4-6	6-8	8-10	10-12	12-14
No. of workers	10	-	25	50	-	15	10

12. From data below, compute Karl Pearson's coefficient of correlation. Also test the significance of correlation.

X	Y
250	280
255	282
280	290
290	295
295	298
300	295

13. From data below, obtain age of car whose maintenance cost is Rs. 3750.

Age of car	Maintenance cost (00)
2	10
4	20
6	25
8	30

14. Ram and Co. manufactures Chrome glass lamp manually. It requires 40 labor hours to complete on with standard deviation of 10 hours.

i) What is the probability that it will take between 35 and 42 labor hours.
ii) What is the probability that will take more than 48 hours?

15. From data below, a person is selected randomly. Find probability that

i) Selected person is male
ii) Selected person is male given that person belongs to manufacturing department.
iii) Selected person is from production control department
iv) Selected person is from quality control department given that person is female.

Department	Gender		Total
	Male	Female	
Manufacturing	280	220	500
Production control	175	125	300
Quality control	115	85	200
Total	570	430	1000

16. Fit Poisson distribution from following data

X	0	1	2	3	4	5	6	7	8
f	120	56	45	34	24	12	10	8	3

17. From information below, find

	Section A	Section B
A.M.	6	8
Standard deviations	5	40/3
No of students	30	32

i) Combined mean
ii) Combined Standard deviation

Group C: Attempt Any TWO Questions: [2X10=20]

18. Lives of two models of refrigerator in a recent survey are:

Life (No. of years)	No of refrigerators	
	Model A	Model B
0-2	5	2
2-4	16	7
4-6	13	12
6-8	7	19
8-10	5	9
10-12	4	1

Which model has greater uniformity

19. From data below find limits and range of middle 40 percent observations

X	10-20	20-30	30-40	40-50	50-60	60-70
f	5	12	18	12	10	4

20. A family has four children. Find probability that there are

a) Exactly four boys
b) no boys
c) At least one boy
d) at least two boys
e) More than two boys